

Glossary of Terms

CBE 30235

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1 Lecture 1 Terms

Capacity Factor The ratio of the actual electrical energy output of a power plant over a given period to the maximum possible electrical energy output over that same period.

Curtailment The deliberate reduction in output below what could have been produced (usually renewables) in order to balance energy supply and demand or due to transmission constraints.

Duck Curve A graph of power production over the course of a day that shows the timing imbalance between peak demand and renewable energy production, specifically the drop in net load during midday and the sharp ramp-up at sunset.

Dunkelflaute A German term ("dark wind lull" or "dark doldrums") referring to periods of time in which little to no energy can be generated using wind and solar power, typically lasting several days.

Firm Power Power generation capacity that can be controlled by the grid operator and is guaranteed to be available at any time, for any duration, regardless of weather conditions (e.g., nuclear, hydro, fossil).

Inertia (Grid) The kinetic energy stored in the rotating mass of large synchronous generators (like steam turbines) that resists changes in grid frequency, providing a buffer against sudden power imbalances.

LCOE (Levelized Cost of Electricity) A measure of the average net present cost of electricity generation for a generating plant over its lifetime, often broken down into Plant-Level LCOE and System-Level LCOE.

Rankine Cycle A thermodynamic cycle which converts heat into mechanical work (and then electricity), commonly used in coal, nuclear, and natural gas power plants.

Swing Equation An electromechanical equation describing the relative motion of the rotor of a synchronous machine, relating the change in frequency ($\frac{df}{dt}$) to the imbalance between mechanical power input and electrical power output.

Terawatt-hour (TWh) A unit of energy equal to 10^{12} watt-hours; commonly used to measure annual national electricity consumption.

2 Lecture 2 Terms

Atomic Mass Unit (u) A unit of mass used to express atomic and molecular weights, defined as one-twelfth of the mass of an unbound neutral atom of carbon-12. ($1 \text{ u} \approx 931.5 \text{ MeV}/c^2$).

Baryon A composite subatomic particle made up of three quarks (e.g., protons and neutrons). Baryons are a subtype of Hadrons.

Boson A particle with integer spin ($0, 1, 2, \dots$) that does not obey the Pauli Exclusion Principle. Bosons include force carriers like photons and gluons.

Electron-volt (eV) A unit of energy equal to the amount of kinetic energy gained by a single electron accelerating from rest through an electric potential difference of one volt. ($1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}$).

Fermion A particle with half-integer spin ($1/2, 3/2, \dots$) that follows the Pauli Exclusion Principle. Includes quarks, leptons, protons, and neutrons.

Hadron Any composite particle made of quarks held together by the strong force. Hadrons are categorized into Baryons (stable) and Mesons (unstable).

Isobar Nuclides which have the same mass number (A) but different atomic numbers (Z).

Isotone Nuclides which have the same neutron number (N) but different atomic numbers (Z).

Isotope Nuclides which have the same atomic number (Z) but different neutron numbers (N).

Lepton An elementary particle of half-integer spin that does not undergo strong interactions. The most prominent examples are the electron and the neutrino.

Pauli Exclusion Principle A quantum mechanical principle stating that two or more identical fermions cannot occupy the same quantum state within a quantum system simultaneously.

Strong Force The fundamental interaction that confines quarks into hadrons and binds protons and neutrons together in the atomic nucleus.

3 Lecture 3 Terms

Binding Energy The energy required to separate a nucleus into its constituent protons and neutrons; equivalently, the energy released when nucleons bind together.

Chart of the Nuclides A two-dimensional graph in which the X-axis represents the number of neutrons and the Y-axis represents the number of protons, mapping all known stable and radioactive isotopes.

Gamma Ray (γ) High-energy electromagnetic radiation (photons) emitted by an atomic nucleus during a transition from a higher energy state to a lower energy state.

Generator ($^{99}\text{Mo}/^{99m}\text{Tc}$) A device used in nuclear medicine to extract the short-lived isotope Technetium-99m from decaying Molybdenum-99.

Isomeric Transition (IT) A radioactive decay process in which a metastable nucleus releases energy via gamma emission to reach a lower energy state without changing its proton or neutron number.

Magic Numbers Specific numbers of nucleons (2, 8, 20, 28, 50, 82, 126) that correspond to complete shells within the atomic nucleus, resulting in higher stability.

Metastable State (m) A nuclear isomer that has a half-life longer than 10^{-9} seconds, often designated with an "m" (e.g., ^{99m}Tc).

Mass Defect The difference between the sum of the masses of the individual components of a nucleus (protons and neutrons) and the actual mass of the nucleus.

Technetium-99m A metastable nuclear isomer used in tens of millions of medical diagnostic procedures annually due to its ideal gamma energy (140 keV) and short physical half-life (6 hours).

Valley of Stability The region in the Chart of the Nuclides where stable isotopes are found; isotopes outside this valley are radioactive and decay toward it.

4 Lecture 4 Terms

Activity (A) The rate at which a sample of radioactive material decays; defined as the number of disintegrations per second. ($A = \lambda N$).

Beta Particle (β) A high-energy, high-speed electron or positron emitted by the radioactive decay of an atomic nucleus.

Becquerel (Bq) The SI unit of radioactivity, defined as one disintegration per second.

Bremsstrahlung "Braking radiation"; electromagnetic radiation (X-rays) produced by the deceleration of a charged particle (like a beta particle) when deflected by another charged particle (like an atomic nucleus).

Curie (Ci) A non-SI unit of radioactivity roughly equivalent to the activity of 1 gram of Radium-226 ($1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}$).

Electron Capture (EC) A decay mode for proton-rich nuclei where an orbital electron is captured by the nucleus, converting a proton into a neutron. It competes with positron emission.

Half-Life ($T_{1/2}$) The time required for a quantity to reduce to half of its initial value. In nuclear physics, $T_{1/2} = \ln(2)/\lambda$.

Mean Life (τ) The average time a radioactive particle exists before decaying. Mathematically, the reciprocal of the decay constant ($\tau = 1/\lambda$).

Neutrino (ν) A fermion that interacts only via the weak subatomic force and gravity. It is electrically neutral and has a very small, non-zero mass. Essential for conserving energy and momentum in beta decay.

Positron Emission (β^+) A type of radioactive decay in which a proton inside a nucleus is converted into a neutron while releasing a positron and an electron neutrino.

5 Lecture 5 Terms

Atmospheric Spallation A nuclear reaction in which a high-energy particle (cosmic ray) strikes an atomic nucleus (like atmospheric Nitrogen), shattering it into smaller component particles and releasing a shower of neutrons.

Cosmogenic Radionuclide Rare isotopes (e.g., ^{14}C , ^{10}Be) created when high-energy cosmic rays interact with an atomic nucleus in the atmosphere or on the surface of Earth.

Coulomb Barrier The energy barrier due to electrostatic interaction that two nuclei must overcome to get close enough to undergo a nuclear reaction. Solar wind protons generally lack the energy to overcome this barrier.

Coronal Mass Ejection (CME) A significant release of plasma and accompanying magnetic field from the solar corona. While they cause magnetic storms, the particles are generally too low-energy to cause nuclear transmutations.

Geomagnetically Induced Current (GIC) Quasi-DC electrical currents induced in power lines and pipelines by rapid changes in the Earth's magnetic field during a solar storm (Space Weather).

Galactic Cosmic Ray (GCR) High-energy particles (mostly protons) originating from outside the solar system (e.g., supernovae) capable of penetrating the Earth's magnetosphere and triggering spallation.

Heliopause The theoretical boundary where the Sun's solar wind is stopped by the interstellar medium; the outer edge of the Sun's magnetic influence.

Single Event Upset (SEU) A change of state caused by a single ionizing particle (like a cosmic ray secondary neutron) striking a sensitive node in a microelectronic device, such as a memory bit flip.

Solar Wind A stream of charged particles (plasma) released from the upper atmosphere of the Sun, consisting mostly of electrons, protons, and alpha particles with kinetic energies in the keV range.

6 Lecture 6 Terms

Accelerator Mass Spectrometry (AMS) A technique that separates isotopes by mass and charge using high-energy acceleration. It allows for the counting of individual ^{14}C atoms, requiring only milligram-sized samples and removing Nitrogen-14 interference.

Bomb Pulse The sudden, massive increase in atmospheric ^{14}C concentration (peaking in 1963) caused by neutron activation during atmospheric thermonuclear weapons testing.

Curve of Knowns The initial validation plot created by Willard Libby, comparing the measured radiocarbon age of artifacts against their known historical age (e.g., Egyptian timber) to prove the method's accuracy.

Dendrochronology The scientific method of dating tree rings. It provides the absolute timeline (Master Chronology) used to calibrate radiocarbon dates against calendar years.

De Vries Effects ("Wiggles") Fluctuations or "wiggles" in the radiocarbon calibration curve caused by variations in solar activity. These plateaus can lead to large uncertainties where a single radiocarbon age corresponds to multiple calendar dates.

Modern Standard (pMC) The baseline activity for radiocarbon dating, defined as 95% of the activity of Oxalic Acid I (normalized to 1950 AD), representing the pre-industrial atmosphere (approx. 15.3 dpm/g).

Poisson Statistics The probability theory governing random events like radioactive decay. It dictates that the standard deviation of a measurement is the square root of the total counts ($\sigma = \sqrt{N}$), defining the precision limit of dating.

Reservoir Effect The phenomenon where different carbon reservoirs (atmosphere, ocean, biosphere) have different ^{14}C concentrations. The ocean acts as a buffer, damping atmospheric fluctuations but also causing marine samples to appear "older" than terrestrial ones.

Suess Effect The dilution of atmospheric ^{14}C levels caused by the combustion of fossil fuels (coal, oil) which contain stable carbon but no radiocarbon, causing modern samples to appear artificially old.

7 Lecture 7 Terms

Alpha Decay A type of radioactive decay where an atomic nucleus emits an alpha particle (two protons and two neutrons), transforming into a new element with a mass number reduced by 4 and atomic number reduced by 2.

Bragg Peak The phenomenon where energetic charged particles (like alphas) deposit most of their energy at the very end of their path. This makes alpha emitters extremely dangerous if internalized, as they dump their energy into a small volume of tissue.

Geiger-Nuttall Law An empirical rule relating the short half-life of an alpha emitter to the high energy of the emitted particle. It was later explained by quantum tunneling mechanics.

Linear Energy Transfer (LET) A measure of the energy transferred from ionizing radiation to soft tissue per unit distance. Alpha particles are "High-LET" radiation.

Quantum Tunneling The quantum mechanical phenomenon where a particle passes through a potential energy barrier that it classically could not surmount. This is the mechanism allowing alpha particles to escape the strong nuclear force.

Radon-222 A radioactive, colorless, odorless noble gas produced by the decay of Radium-226. It is the primary cause of environmental radiation exposure for the general public.

Secular Equilibrium A steady-state condition in a radioactive decay chain where the quantity of a radioactive daughter remains constant because its production rate (from a long-lived parent) equals its decay rate.

Sub-Slab Depressurization The standard engineering solution for high indoor radon. It involves installing a fan and vent pipe to create a vacuum beneath a building's foundation, venting soil gases to the outside air before they can enter.

TENORM "Technologically Enhanced Naturally Occurring Radioactive Material." Natural radioactive material that has been concentrated or exposed to the environment by human activities, such as mining or coal combustion.

8 Lecture 8 Terms

Closure Temperature The temperature below which a specific mineral crystal structure "locks in" its isotopes, preventing the diffusion of parent or daughter atoms. For dating, this marks the "time zero" of the clock.

Concordia Diagram A plot used in U-Pb dating. The "Concordia" is the curve of points where the ^{238}U and ^{235}U clocks agree. Points falling off this curve ("Discordant") indicate lead loss or contamination.

Discordia Line A linear regression line drawn through discordant data points on a Concordia diagram. The upper intercept with the Concordia curve indicates the original crystallization age, while the lower intercept indicates the time of a metamorphic event (lead loss).

Isochron A graphical technique used in Rb-Sr and other dating methods to solve for age when the initial amount of daughter isotope is unknown. By plotting ratios of different minerals from the same rock, the age is determined by the slope of the resulting line.

K-Ar Dating A radiometric dating method useful for volcanic rocks. It relies on the branching decay of ^{40}K to ^{40}Ar . Because Argon gas escapes from magma, the clock effectively resets upon eruption/solidification.

Radiogenic Heat Thermal energy produced within the Earth by the radioactive decay of isotopes, primarily ^{238}U , ^{232}Th , and ^{40}K . It accounts for approximately 50% of the Earth's total heat flux.

Zircon (ZrSiO_4) A nesosilicate mineral widely used in U-Pb dating. It is mechanically hard, chemically resistant, and has a high closure temperature. Crucially, it accepts Uranium into its lattice but excludes Lead during formation.

9 Lecture 9 Terms

Barn (b) The standard unit of area for nuclear cross sections. $1 \text{ b} = 10^{-24} \text{ cm}^2$. It was named "barn" because, to nuclear physicists, a nucleus as large as Uranium looked "as big as a barn."

Flux (ϕ) The scalar flux is the total distance traveled by all neutrons in a unit volume per unit time. Defined as $\phi = nv$ (neutron density $\times$ speed), with units of neutrons/cm²-s. It determines the rate of reactions in a reactor.

Macroscopic Cross Section (Σ) The probability of neutron interaction per unit path length of travel in a bulk material. $\Sigma = N\sigma$, where N is the atom density. Units are cm⁻¹.

Mean Free Path (λ) The average distance a neutron travels in a material before undergoing an interaction. It is the reciprocal of the total macroscopic cross section: $\lambda = 1/\Sigma_t$.

Microscopic Cross Section (σ) The effective target area a single nucleus presents to an incident neutron, representing the probability of a specific interaction (scattering, fission, etc.). Units are barns.

Radiative Capture (n, γ) An absorption reaction where the neutron is captured by the nucleus, which then releases the excess excitation energy as a gamma ray. This is the primary mechanism for creating isotopes (e.g., Co-60) and breeding fuel (e.g., Pu-239).

Scattering (Elastic vs. Inelastic) • *Elastic*: KE is conserved in the center-of-mass system; the neutron bounces off. (Moderation).

- *Inelastic*: The neutron strikes the nucleus and excites it; the nucleus later emits a gamma, and the neutron loses significant energy. (Fast neutron slowing).

10 Lecture 10 Terms

1/v Region The low-energy "thermal" region where the neutron absorption cross-section is inversely proportional to the neutron velocity ($\sigma_a \propto 1/v$). This explains why slow neutrons are so efficient at causing fission.

Center of Mass (COM) System A coordinate system that moves with the average velocity of the colliding particles. In nuclear kinematics, analyzing collisions in the COM frame simplifies the math because the total momentum is zero, making calculations of energy transfer straightforward.

Collision Parameter (α) A distinct property of a target nucleus determined by its mass number A . It defines the minimum energy a neutron can retain after a single elastic collision: $E_{min} = \alpha E_{in}$. For Hydrogen, $\alpha = 0$; for heavy elements, $\alpha \approx 1$.

Fast Reactor A nuclear reactor design that utilizes fast neutrons (un-moderated) to sustain the chain reaction. It requires highly enriched fuel and non-moderating coolants (like liquid sodium) but offers the ability to breed fissile material.

Logarithmic Energy Decrement (ξ) The average loss in the natural logarithm of neutron energy per collision. It serves as a figure of merit for moderators; a higher ξ means fewer collisions are required to thermalize a neutron.

Moderator A material with low atomic mass (high ξ) and low absorption cross-section used to slow down fast neutrons to thermal energies. Common examples include light water ($^1\text{H}_2\text{O}$), heavy water (D_2O), and nuclear graphite.

Resonance Region The energy range (typically 1 eV to 100 keV) where the neutron cross-sections of heavy nuclei (like ^{238}U) exhibit sharp peaks (resonances). This is the "Valley of Death" where neutrons are easily lost to non-fission capture.

Thermal Neutron A neutron that has slowed down enough to reach thermal equilibrium with its surroundings (typically ~ 0.025 eV at room temperature).

11 Lecture 11 Terms

Decay Heat The heat released by the radioactive decay of fission products after the fission event itself. Although it accounts for only $\sim 7\%$ of total power, it continues after reactor shutdown, necessitating active cooling to prevent meltdown (e.g., Fukushima).

Fissile Isotope An isotope (like ^{235}U or ^{239}Pu) that can be fissioned by *thermal* (slow) neutrons. This occurs because the binding energy released by neutron capture alone is sufficient to overcome the critical deformation threshold.

Fissionable Isotope An isotope (like ^{238}U) that can undergo fission, but only if struck by a high-energy (*fast*) neutron. The binding energy of capture is insufficient to split the nucleus; the neutron must bring kinetic energy to bridge the gap.

Liquid Drop Model A theoretical model treating the nucleus as a drop of incompressible nuclear fluid. It successfully explains fission as a competition between the disruptive Coulomb force (protons repelling) and the restorative Surface Tension (strong force).

Prompt Neutrons Neutrons emitted almost instantaneously ($< 10^{-14}$ s) during the fission process. They consist of "scission neutrons" (from the neck snap) and "evaporation neutrons" (boiled off by hot fragments).

Reproduction Factor (η) The average number of fission neutrons produced per neutron *absorbed* in the fuel. $\eta = \nu(\sigma_f/\sigma_a)$. This determines the maximum possible breeding ratio or multiplication factor for a given fuel type.

Scission The exact moment when the highly deformed nuclear "neck" snaps, separating the nucleus into two distinct fission fragments.

12 Lecture 12 Terms

Four Factor Formula The equation describing the infinite multiplication factor in a thermal reactor: $k_\infty = \eta\epsilon pf$. It decomposes the neutron life cycle into four distinct physical probabilities.

Heterogeneous Reactor A reactor design where the fuel and moderator are physically separated (e.g., solid fuel rods in liquid water). This geometry is essential for using low-enriched uranium because it dramatically increases the Resonance Escape Probability (p).

Infinite Multiplication Factor (k_∞) The ratio of neutrons in one generation to the previous generation in an infinite medium (ignoring leakage). If $k_\infty = 1$, the population is stable.

Oklo Reactor A set of natural nuclear fission reactors that operated in Gabon, West Africa, approximately 1.7 billion years ago. They are the only known instance of naturally occurring critical chain reactions on Earth, made possible by higher ancient concentrations of ^{235}U .

Resonance Escape Probability (p) The probability that a neutron will slow down from fission energy to thermal energy without being captured by the resonance peaks of ^{238}U . "Lumping" fuel into rods increases p .

Thermal Utilization Factor (f) The ratio of thermal neutrons absorbed in the fuel to the total thermal neutrons absorbed in the entire unit cell (fuel + moderator + cladding). "Lumping" fuel decreases f due to self-shielding.

Unit Cell The fundamental, repeating geometric component of a nuclear reactor core (typically a single fuel rod surrounded by its associated coolant/moderator channel). Reactor physics calculations are often performed on a single unit cell and then scaled up.

13 Lecture 13 Terms

Buckling (B^2) A measure of the "bending" of the neutron flux profile. "Material Buckling" (B_m^2) depends on fuel properties (k_∞) and describes what the fuel *can* do. "Geometric Buckling" (B_g^2) depends on the reactor size and describes what the core *needs* to be critical. For criticality, $B_m^2 = B_g^2$.

Chicago Pile-1 (CP-1) The world's first artificial nuclear reactor, built by Enrico Fermi's team in 1942. It used natural uranium fuel and graphite moderation to achieve criticality ($k = 1.0006$) at 0.5 Watts of power.

Diffusion Coefficient (D) In nuclear engineering, a parameter representing the ability of neutrons to travel through a medium without scattering back. Unlike chemical diffusivity, it has units of **Length (cm)** because it is derived from the transport mean free path ($\lambda_{tr}/3$).

Fick's Law The approximation used to model neutron transport in high-scattering media. It states that the net neutron current is proportional to the negative gradient of the flux: $J = -D\nabla\phi$.

Homogenization The approximation method where complex heterogeneous lattices (fuel rods + water + steel) are treated as a single uniform material with "effective" cross-sections. This is valid because the reactor scale is much larger than the neutron mean free path.

Neutron Current (J) A vector quantity representing the net number of neutrons crossing a unit surface area per unit time. It describes the *flow* or leakage of neutrons.

SCRAM The emergency shutdown of a nuclear reactor. While often attributed to the "Safety Control Rod Axe Man" from CP-1, it technically refers to the rapid insertion of negative reactivity (control rods) to immediately halt the fission chain reaction.

14 Lecture 14 Terms

Autocatalysis A chemical reaction where a product also serves as a catalyst for the reaction (e.g., $A + B \rightarrow 2A$). Nuclear fission is a physical form of autocatalysis (neutrons produce more neutrons), and its governing diffusion equation is identical to that of chemical autocatalysis.

Geometric Buckling (B_g^2) A parameter that describes the neutron leakage of a specific reactor shape and size. It is mathematically the lowest eigenvalue of the Helmholtz equation ($\nabla^2\phi + B^2\phi = 0$) for the system's geometry. For a reactor to be critical, the geometric buckling (demand) must equal the material buckling (supply).

Reflector A layer of material (usually a moderator like water, beryllium, or graphite) surrounding the reactor core. It scatters escaping neutrons back into the fuel, reducing leakage and improving the neutron economy.

Reflector Savings (δ) The reduction in the critical dimension of a reactor core made possible by the addition of a reflector. Mathematically, the reflected core behaves like a bare core of size $L + 2\delta$.

Robin Boundary Condition A type of boundary condition that specifies a linear combination of the function and its derivative on the boundary ($a\phi + b\phi' = 0$). In reactor physics, this describes the interface between the core and a reflector, analogous to the convective heat transfer boundary condition in thermodynamics.

Vacuum Boundary Condition The approximation that no neutrons enter the reactor from the outside vacuum. This mathematically implies that the neutron flux goes to zero at a specific distance slightly beyond the physical edge of the reactor (the Extrapolated Distance).

15 Lecture 15 Terms

Delayed Neutrons A small fraction ($\beta \approx 0.65\%$) of fission neutrons that are emitted seconds or minutes after the fission event. They originate from the decay of fission products (Precursors) rather than the fission itself. They are the essential physical phenomenon that makes nuclear reactor control possible.

Fermi Age (τ_T) Despite the name, this parameter has units of Area (cm^2). It represents roughly 1/6 of the mean square distance a fast neutron travels while slowing down to thermal energy. It is the "fast" analog to the diffusion length squared (L^2).

Six-Factor Formula The extension of the Four-Factor Formula to finite reactors. It includes two new terms: the Fast Non-Leakage Probability (P_{FNL}) and the Thermal Non-Leakage Probability (P_{TNL}). $k_{eff} = \epsilon p \eta f P_{FNL} P_{TNL}$.

Prompt Neutrons Neutrons emitted immediately at the moment of scission ($< 10^{-14}$ s). In a prompt-critical excursion (like the Cecil Kelley accident), power rises on a millisecond timescale driven purely by these neutrons.

Reactivity (ρ) A measure of the departure of a reactor from criticality. Defined as $\rho = (k_{eff} - 1)/k_{eff}$. It is often measured in units of "dollars" and "cents," relative to the delayed neutron fraction β .

Reactor Period (T) The time required for the neutron population (and reactor power) to change by a factor of e (≈ 2.718). In the prompt approximation, $T = l/(k - 1)$. A short period means rapid, potentially dangerous power changes.

16 Lecture 16 Terms

Dollar (\$) A unit of reactivity defined relative to the delayed neutron fraction. One dollar (\$1.00) of reactivity is equal to β . A reactor with \$1.00 of excess reactivity is Prompt Critical.

Inhour Equation The equation relating the reactivity (ρ) of a reactor to its stable period (T). It demonstrates that for small reactivity insertions ($\rho < \beta$), the reactor period is determined by the long decay times of the precursors, not the short lifetime of the prompt neutrons.

Point Kinetics A simplified model of reactor dynamics that ignores spatial variations in neutron flux. It assumes the "shape" of the flux remains constant while the "amplitude" changes over time. It consists of coupled differential equations for the neutron density (n) and precursor concentration (C).

Precursors Fission products (such as ^{87}Br or ^{137}I) that undergo beta decay and subsequently emit a neutron. These atoms act as a "storage bank" for neutrons, releasing them seconds or minutes after the fission event, which slows down the overall reactor response.

Prompt Critical The state where a reactor is critical ($k = 1$) based solely on prompt neutrons, without needing the delayed neutrons. This occurs when $\rho \geq \beta$. In this state, the reactor period becomes extremely short (milliseconds), usually leading to a destructive power excursion.

17 Lecture 17 Terms

Doppler Broadening The physical phenomenon where the thermal motion of nuclei "blurs" the energy of incoming neutrons relative to the nucleus. In U-238, this widens the resonance capture peaks, increasing neutron absorption as temperature rises. It provides the essential instantaneous negative feedback that stabilizes a reactor.

Moderator Temperature Coefficient (α_{mod}) The change in reactivity per degree change in moderator temperature. In a properly designed PWR (under-moderated), this is negative: as water heats and expands, moderation becomes less efficient, and the chain reaction slows down.

Positive Void Coefficient A dangerous design characteristic (found in the Chernobyl RBMK design) where a reduction in coolant density (boiling/voids) leads to an *increase* in reactivity. This creates a runaway positive feedback loop: Power $\rightarrow$ Boiling $\rightarrow$ Reactivity $\uparrow \rightarrow$ More Power.

Power Defect The total amount of negative reactivity inserted by the feedback mechanisms (Doppler + Moderator) as the reactor goes from zero power (hot standby) to full operating power. The control rods must add enough positive reactivity to "pay" for this defect to maintain criticality at full power.

Prompt Criticality The threshold where the reactor becomes critical on prompt neutrons alone ($\rho \geq \beta$). While typically associated with rod withdrawal accidents, it can also be triggered by a "Cold Water Accident" (rapid cooling of the moderator) if the negative temperature coefficient is strong enough.

18 Lecture 18 Terms

Burnout (Nuclear) The process by which a high-cross-section isotope (like ^{135}Xe) is removed from the reactor by capturing a neutron and transmuting into a lower-cross-section isotope. For Xenon, burnout is the dominant removal mechanism during full-power operation.

Iodine Pit The temporary surge in Xenon-135 concentration that occurs after a reactor is shut down. Because the "burnout" mechanism stops (no flux) but the "production" mechanism continues (decay of existing Iodine-135), the poison level rises to a peak roughly 10 hours after shutdown, making restart difficult or impossible ("Xenon Precluded").

Neutron Poison A material with a large neutron absorption cross-section that absorbs neutrons without causing fission, thereby decreasing reactivity. Some are added intentionally (Control Rods, Burnable Absorbers), while others are unwanted fission products (Xenon, Samarium).

Promethium-149 The radioactive fission product precursor to Samarium-149. Its decay after shutdown leads to the "Samarium Peak."

Samarium-149 A stable fission product poison. Unlike Xenon, it does not decay. Once formed (mostly after shutdown), it remains in the core until the reactor is restarted and the poison is "burned out" by neutron flux.

Xenon-135 The most significant fission product poison in thermal reactors, with a thermal absorption cross-section of 2.6×10^6 barns. Its dynamic behavior dictates the timescales for reactor restart and load-following operations.

Xenon Oscillation A spatial instability in large reactors where power shifts back and forth between different regions of the core (like water sloshing in a bathtub) due to the delay between flux changes and Xenon production/burnout.

19 Lecture 19 Terms

Differential Rod Worth ($d\rho/dz$) The amount of reactivity added per unit of control rod movement (e.g., pcm per inch). It follows the square of the neutron flux (ϕ^2), resulting in a peaked distribution that is highest in the center of the core and near zero at the boundaries.

Integral Rod Worth The cumulative reactivity inserted by a control rod as it is withdrawn from zero to a specific position z . The plot of Integral Worth vs. Position forms the characteristic "S-Curve" (Sigmoid shape), which dictates how operators respond to reactivity changes.

Perturbation Theory A mathematical method used to estimate the effect of small changes (like inserting a thin absorber rod) on the reactor system. It demonstrates that the reactivity worth of an absorber is proportional to the product of the Flux (ϕ) and the Adjoint Flux ($\phi^\dagger$, or "Importance").

Rod Ejection Accident A Design Basis Accident (DBA) where a failure of the control rod drive housing causes the high-pressure coolant to shoot a control rod out of the core like a projectile. The safety limits (Rod Insertion Limits) are calculated to ensure that even if the highest-worth rod is ejected, the reactor does not go Prompt Critical.

SL-1 Accident A fatal nuclear accident in 1961 caused by the manual withdrawal of a central control rod. It highlighted the danger of the "steep" portion of the integral rod worth curve and led to regulations requiring that reactors remain subcritical even with the single most reactive rod fully stuck out of the core (The Stuck Rod Criterion).

20 Lecture 20 Terms

Chemical Shim The practice of dissolving a neutron absorber (typically Boric Acid) directly into the reactor coolant. This provides global, uniform reactivity control, allowing control rods to be kept fully withdrawn to minimize flux distortion and power peaking.

Flux Flattening The engineering process of modifying the reactor design (via reflectors, variable enrichment, or poisons) to make the neutron flux distribution as uniform as possible. This lowers the Peaking Factor, allowing the reactor to be run at a higher total power without melting the central fuel pins.

Gadolinium (Burnable Poison) A simplified method of flux shaping where a high-cross-section material (Gadolinium) is mixed directly into the fuel pellets of fresh assemblies. It suppresses the high power density of new fuel and burns out as the fuel depletes, keeping the power distribution stable over time.

Peaking Factor (F_q) The ratio of the maximum power density in the core to the average power density (P_{max}/P_{avg}). A high F_q indicates inefficient fuel usage and thermal bottlenecks. Minimizing this ratio is a primary goal of core design.

Xenon Oscillation A spatial instability occurring in large reactors where power shifts between different regions of the core (e.g., top vs. bottom) due to the time-delayed interaction between Iodine decay (production) and Xenon burnout (removal). Operators must carefully manage control rods to dampen these waves.

Zone Loading A fuel management strategy where fuel assemblies with different U-235 enrichments are placed in specific patterns. Typically, higher enrichment fuel is placed at the core periphery (to boost low flux) and lower enrichment at the center (to suppress high flux), flattening the overall power shape.

21 Lecture 21 Terms

Group Constants The averaged nuclear cross-sections ($D_g, \Sigma_{ag}, \Sigma_{fg}$) used in multigroup diffusion theory. They are obtained by weighing the energy-dependent cross-sections by the neutron flux spectrum within that energy group.

Reflector Hump A phenomenon observed in the thermal neutron flux profile near the interface of a reactor core and a moderator (reflector). Fast neutrons leak into the reflector, slow down, and accumulate because they are not absorbed. This causes a local peak in thermal flux, often exceeding the flux at the core boundary.

Removal Cross Section ($\Sigma_{1 \rightarrow 2}$) A macroscopic cross-section representing the probability that a fast neutron will scatter out of the fast group (Group 1) and enter the thermal group (Group 2). It effectively acts as the "absorption" term for the fast group and the "source" term for the thermal group.

Two-Group Approximation A simplified model of neutron transport where the energy spectrum is divided into two discrete bins: Fast (born from fission) and Thermal (in equilibrium with the moderator). It is the minimum level of complexity required to accurately model reflected reactors and criticality in water-moderated systems.

22 Lecture 22 Terms

TBA Glossary terms for this lecture will be compiled based on the specific technical topics covered by Captain Ryan.

23 Lecture 23 Terms (Historical Context)

CRUD Acronym for **C**halk **R**iver **U**nidentified **D**eposit. Originally describing radioactive corrosion products found in Canadian test loops, it is now the standard industry term for activated corrosion products (rust) that circulate in the primary coolant and deposit on reactor surfaces.

Polaris The first submarine-launched ballistic missile (SLBM) system. Unlike Regulus missiles (which required the sub to surface), Polaris could be launched while submerged, creating a survivable second-strike nuclear deterrent.

Admiral H.G. Rickover The Director of the Naval Reactors Branch. Known for his "persuasive personality," he shifted the AEC's focus from breeding reactors to submarine propulsion and established the rigorous engineering standards (the "philosophy of worry") that define the naval nuclear program.

Sodium-Cooled Reactor A reactor design using liquid sodium metal as the coolant. Used in the USS *Seawolf* (SSN-575). While it offers high thermal efficiency, it was abandoned for naval use due to maintenance difficulties (high gamma radiation from Sodium-24) and the logistical benefits of water.

USS Nautilus The world's first nuclear-powered ship (SSN-571), launched in 1954. It utilized a Pressurized Water Reactor (PWR) and demonstrated the capability to circumnavigate the globe submerged.

Zirconium A metal chosen for fuel cladding because of its high strength, corrosion resistance, and remarkably low neutron absorption cross-section. Its industrial development was forced rapidly by the Naval Reactors program in the late 1940s.